

Module 07: REST API Development

Duration: 5-7 days | **Level:** Intermediate

API Design Principles

Method	URI	Action
GET	/api/tasks	List tasks
POST	/api/tasks	Create task
GET	/api/tasks/{id}	Show task
PUT/PATCH	/api/tasks/{id}	Update task
DELETE	/api/tasks/{id}	Delete task

Use proper HTTP status codes: 200, 201, 204, 401, 403, 404, 422.

API Resources

Transform models to consistent JSON:

```
class TaskResource extends JsonResource
{
    public function toArray(Request $request): array
    {
        return [
            'id' => $this->id,
            'title' => $this->title,
            'status' => $this->status->value,
            'project' => new ProjectResource($this->whenLoaded('project')),
        ];
    }
}
```

Controller:

```
return TaskResource::collection($tasks); // Paginated list
return new TaskResource($task);         // Single item
```

Sanctum Token Auth

```
composer require laravel/sanctum
php artisan vendor:publish --provider="Laravel\Sanctum\SanctumServiceProvider"
php artisan migrate
```

Generate token:

```
$token = $user->createToken('mobile-app')->plainTextToken;
// Returns: "1|abc123..."
```

API request:

```
curl -H "Authorization: Bearer 1|abc123..." \
-H "Accept: application/json" \
http://localhost:8000/api/tasks
```

API Versioning

```
// routes/api.php
Route::prefix('v1')->group(function () {
Route::apiResource('tasks', TaskApiController::class);
});
```

Rate Limiting

```
Route::middleware(['auth:sanctum', 'throttle:60,1'])->group(...);
```

TaskForge API Files

- routes/api.php
 - app/Http/Controllers/Api/TaskApiController.php
 - app/Http/Resources/TaskResource.php
 - tests/Feature/Api/TaskApiTest.php
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Exercises

1. Add POST /api/tasks/{task}/complete endpoint.
 2. Create CommentResource and nest comments in task response.
 3. Write API test for 403 when accessing another user's task.
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Next Module

-> [Module 08: Frontend Assets](#)